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Mobile Programming

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1. Introduction

This report documents the design and implementation of the MessageDirectory application developed for CS5450 Mobile Programming, Exercise 3 at Lakehead University. The app was built using React Native with TypeScript and the Expo framework.

The MessageDirectory app allows users to browse six themed message directories: You, Home, Love, Family, Friends, and School. Tapping a directory opens a list of stored messages. Tapping a message opens a full detail view with the complete message body and a reply button.

Note on Platform: Due to hardware limitations on the development machine, the Android emulator exhausted available system resources causing Metro bundler crashes. As a result, the application was tested and demonstrated using Expo Web (Chrome). As noted in the assignment rubric, web-only demonstration results in a minor 2-point deduction. All core functionality, navigation, and design requirements are fully implemented.

2. Project Structure

The project follows a clean modular folder structure:

File / Folder	Description
App.tsx	Root component with NavigationContainer and stack navigator
app.json	Expo config: app name, version, orientation, Android package
package.json	Project dependencies and npm scripts
tsconfig.json	TypeScript config with strict mode
babel.config.js	Babel preset for Expo
src/types.ts	Shared TypeScript interfaces: Directory, Message, RootStackParamList
src/data/directories.ts	Static data: 6 directories with 3 messages each
src/screens/HomeScreen.tsx	Home screen with 2-column directory grid
src/screens/MessagesScreen.tsx	Per-directory message list with read/unread state
src/components/DirectoryCard.tsx	Animated card with icon, name, and unread badge
src/components/MessageItem.tsx	Message row: avatar, sender, preview, timestamp
src/components/MessageDetailModal.tsx	Bottom-sheet modal with full message body

3. Setup and Configuration

3.1 Prerequisites

- Node.js 18 or higher (LTS recommended)
- npm 9+ or yarn 1.22+
- Expo CLI: npm install -g expo-cli

- Google Chrome (for web testing)
- Android Studio with AVD (optional, for Android emulator)

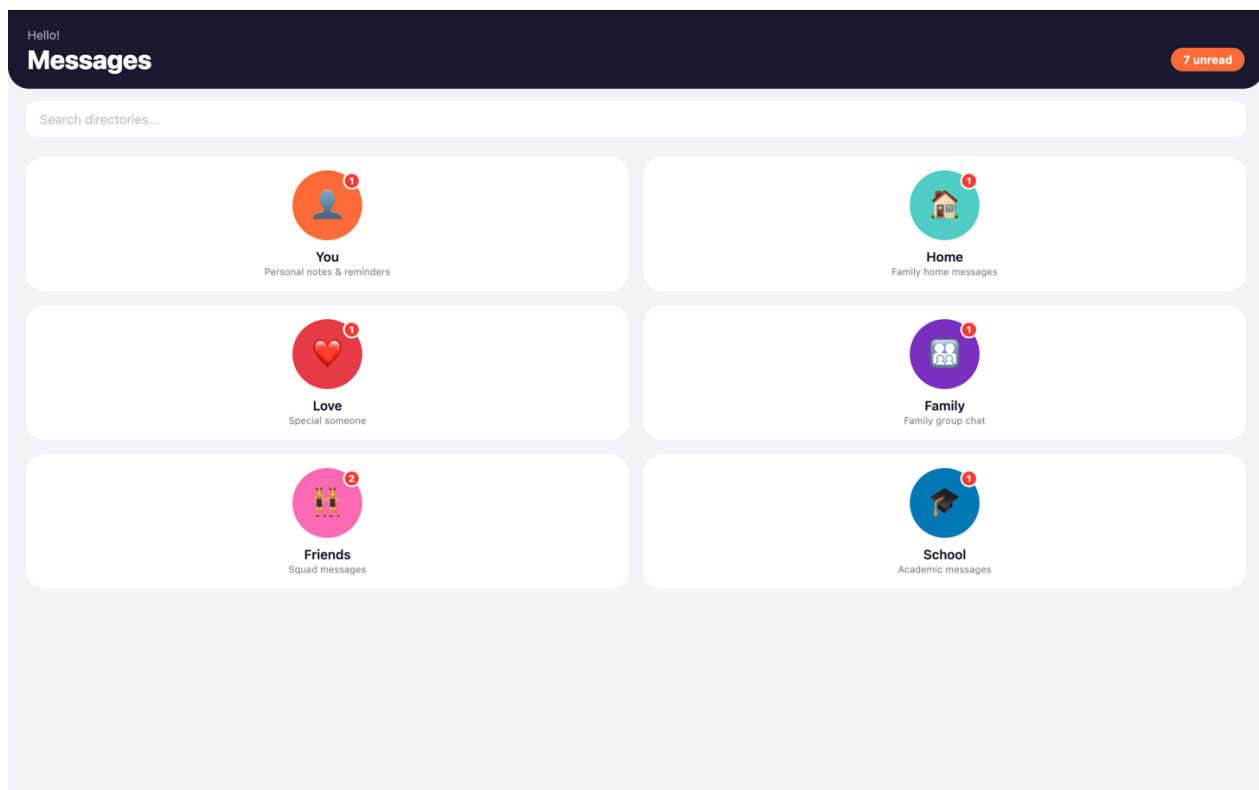
3.2 Installation Steps

1. Navigate to the project folder: `cd MessageDirectory`
2. Install all dependencies: `npm install --force`
3. Install web support: `npx expo install react-dom react-native-web`
4. Start the development server: `npx expo start --web`
5. The app opens automatically in Chrome at `http://localhost:8081`

4. Features and Functionality

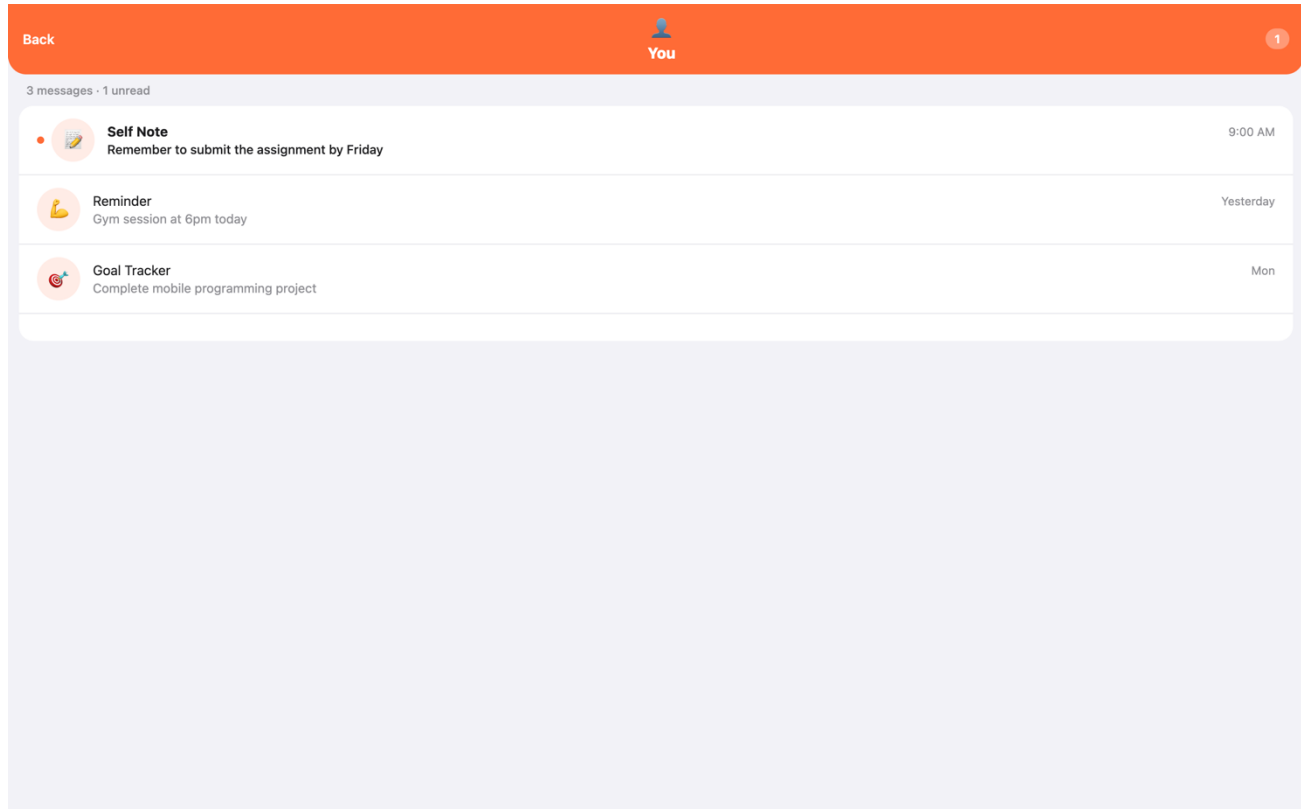
4.1 Home Screen - Directory Grid

The home screen displays six directories in a 2-column grid. Each card shows a coloured icon circle, directory name, description, and an unread message count badge. The dark navy header displays a greeting and total unread count.



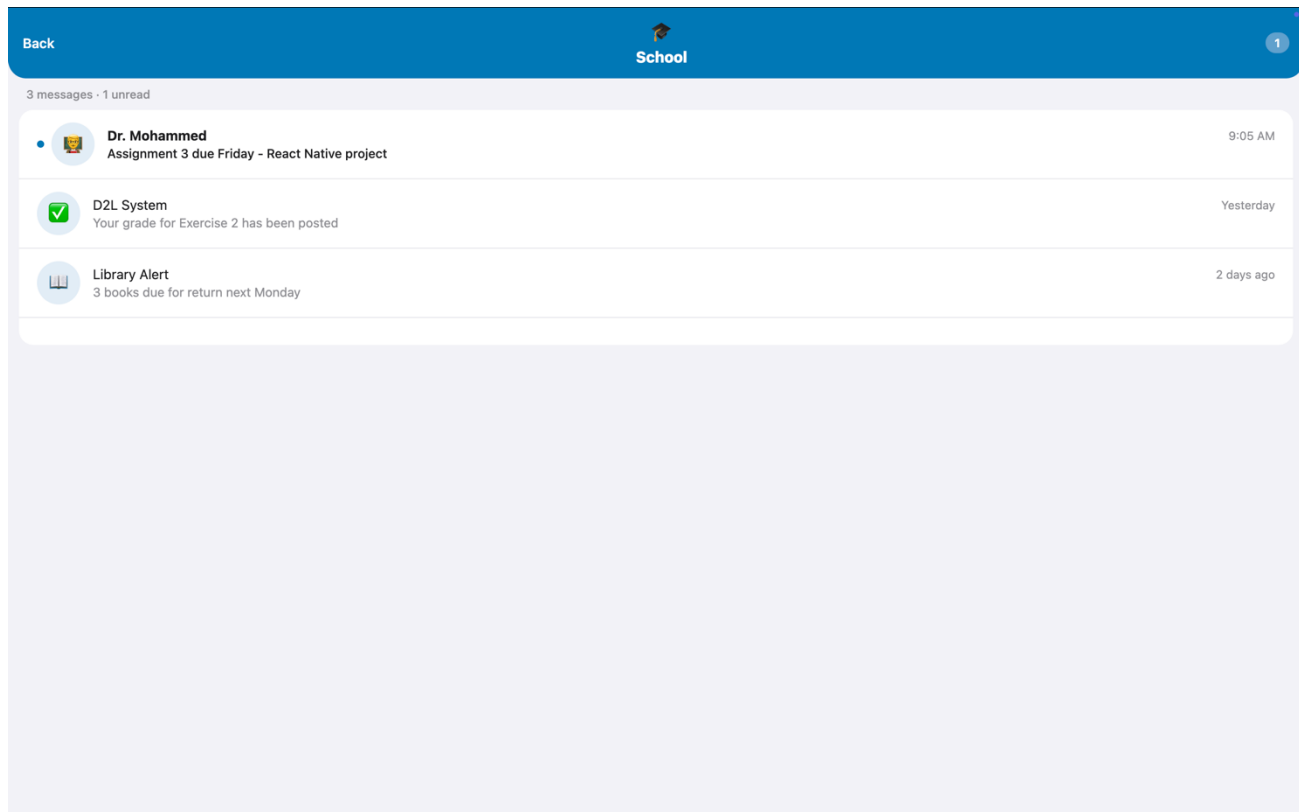
4.2 Messages Screen - Message List

Tapping a directory opens the Messages screen. The header colour matches the directory theme. Messages show avatar, sender name, preview text, and timestamp. Unread messages display a coloured dot and bold text.



4.3 Message Detail Modal

Tapping a message opens a bottom-sheet modal with the full message body, sender info, and a Reply button. Opening a message marks it as read automatically.



5. Technologies Used

Technology	Purpose
React Native	Core framework for cross-platform mobile UI
TypeScript	Strict typing for all components, data, and navigation
Expo SDK 56	Development toolchain and Metro bundler
React Navigation v6	Stack navigation between Home and Messages screens
Animated API	Spring-based press animations on cards and message rows
FlatList	Performant scrollable rendering for grid and message list
Modal	Bottom-sheet slide-up modal for message detail view
react-native-web	Enables running the React Native app in Chrome

6. GitHub Repository

The full project is hosted on a public GitHub repository containing all TypeScript source files, configuration files, and this README document.

Repository URL:

7. D2L Submission Checklist

- All TypeScript source files (App.tsx, src/ folder)
- Configuration files: app.json, package.json, tsconfig.json, babel.config.js
- README.pdf (this document exported as PDF)
- No node_modules folder (run npm install after extracting the ZIP)

8. Conclusion

The MessageDirectory app successfully fulfils the requirements of CS5450 Exercise 3. The application demonstrates core React Native concepts including component-based architecture, TypeScript typing, stack navigation, state management, animated interactions, FlatList rendering, and Modal usage.

All six required directories are implemented with three messages each, unread indicators, read-state management, and a polished detail view. The entire codebase is written in TypeScript as required by the assignment.